Essential in NX NASTRAN (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	Introduction to Finite Element Analysis Areas of application Theories behind FEA FEM Vs. FEA About Pre-Processing, Solver, Post-Processor Introduction to NX Nastran Overview on Advanced Simulation About Simulation File Structure Advanced Simulation Workflow Simulation Navigator Understanding geometry abstraction process About Polygon geometry Fillet Identification Process **Model Cleanup commands:** Auto Heal Geometry, Split Edge, Split face Merge edge, Merge face Stitch Edge, Unstitch edge Collapse edge, Face repair, Reset	Theory/Demo	2.0Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0Hrs
3	Session 3	About geometry idealization process Comparison of geometry idealization and abstraction **Model Preparation commands:** Idealize geometry, Defeature geometry split body, mid-surface user defined method, sew, and divide face. Overview on materials Units and Dimension, Unit Management, Units Manager, Unit Converter Simulation Co-ordinate systems, Nodal coordinate systems Meshing overview, Mesh points, 3D Tetrahedral Mesh, 3D Swept Mesh, Solid from shell Mesh Overview on 2D meshing, Understanding free mapped meshes and creation 2D mapped meshing, 2D Dependent mesh Local Refinement for associative 2D meshes, surface coat of 2D elements 1D meshing, defining beam cross section and orientation 0D meshing and controlling mesh density on edges & faces	Theory/Demo	2.0Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0Hrs
5	Session 5	Creating and applying physical property tables. Defining element and mesh attributes. Material Orientation, Determining Nastran 2D element material orientation Determining Nastran 3D element material orientation, Activity Comprehensive check, Model setup, Element Shapes, Element outlines, Nodes 2D Element Normal, 2D Element Normal by seed, Node proximity to CAD geometry Element Material Orientation, CAE Model consistency, Element Shapes Threshold values. Boundary Condition, Creating loads, Creating constraints, Activity Solving the model, Analysis job monitor, Batch Solving, Checking analysis quality NX Nastran Solution Monitor, Solution information monitor	Theory/Demo	2.0Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0Hrs
7	Session 7	Introduction to Linear Static Analysis. Analysis on 1D element and over link component Introduction to Contact analysis and understanding contact pressure between shaft & wheel Overview on Post processing ,Importing results, Post View Master post view and multiple viewports Managing display of deformed models, Contour plots Post View Templates, Post View Layouts, Overlay Plots Combining load cases, Animation, Identify, Generating Reports, Exporting Reports, Activity Importing Finite Element model data, Exporting Simulation file, Activity	Theory/Demo	2.0Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0Hrs

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9	Session 9	Introduction to gluing and understanding natural frequency in a golf club assembly Introduction to Modal Analysis and analyzing mode shapes in a speaker cabinet with & without connection	Theory/Demo	2.0Hrs
10	Session 10	Exercises included for Practice	Practicals	2.0Hrs
11	Session 11	Introduction to Response Simulation and FE model setup Advanced Post Processing in Response Simulation. To analysis events such as response, transmissibility, transient, random analysis	Theory/Demo	2.0Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0Hrs
13	Session 13	Introduction to thermal analysis and performing thermal & structural analysis	Theory/Demo	2.0Hrs
14	Session 14	Exercises included for Practice	Practicals	2.0Hrs
15	Session 15	Introduction to Nonlinear analysis and studying geometric nonlinearity	Theory/Demo	2.0Hrs
16	Session 16	Exercises included for Practice	Practicals	2.0Hrs
17	Session 17	Introduction to Assembly FEM, Assembly FEM Vs. multi body FEM, Working with sub-assemblies Finding natural frequency in an assembly by mapping sub-assemblies	Theory/Demo	2.0Hrs
18	Session 18	Exercises included for Practice	Practicals	2.0Hrs
19	Session 19	Introduction to Optimization and study to obtain an optimized design for a cross member	Theory/Demo	2.0Hrs
20	Session 20	Exercises included for Practice	Practicals	2.0Hrs